

MD RAISUL ISLAM RAHAD

Software Engineer — B.Sc. in CSE, MIST (2026)
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Education

- Military Institute of Science and Technology (MIST)** Dhaka, Bangladesh
B.Sc. in Computer Science & Engineering CGPA: 3.43 / 4.00
May 2022 – June 2026
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networking, Software Engineering, Artificial Intelligence, Digital Image Processing, Compiler Design

Skills

- Languages:** C/C++ (proficient), Python (proficient), JavaScript (proficient), Java (familiar), SQL, Bash
- Web & Backend:** React.js, Node.js, Express.js, REST API, JWT, OAuth, WebSocket
- Databases:** PostgreSQL, OracleDB, Firebase, Supabase
- Robotics & Embedded:** ROS, OpenCV, Jetson Xavier NX, ESP32, Teensy, ZED Stereo Camera
- Tools & Platforms:** Git, GitHub, Linux, LaTeX, Cisco Packet Tracer, Flutter

Experience

- Siliconova** Dhaka, Bangladesh
Embedded Software Engineer Oct 2026 – Present
 - Developing embedded software solutions for hardware products, working with low-level firmware, microcontrollers, and real-time systems
- Imperial Trends** Leicester, United Kingdom (UK) – Remote
Software Developer June 2026 – Sep 2026
 - Developed and maintained web applications using React.js, Node.js, and PostgreSQL, implementing RESTful APIs and secure authentication mechanisms
 - Collaborated with cross-functional teams to design and deploy scalable software solutions, improving system performance and user experience
 - Conducted code reviews, implemented CI/CD pipelines, and optimized database queries to enhance application efficiency and reliability
- MIST Mars Rover Society** Dhaka, Bangladesh
Team Lead June 2025 – June 2026
 - Led 50+ members across mechanical, electrical, and software teams to develop and deploy a Mars rover for international competitions
 - Represented MIST Mongol Barota at URC 2026 with a 12-member crew, securing 11th place among 38 finalists worldwide
 - Managed project timelines, technical reviews, and international competition logistics

Software & Communication Co-Lead / Mentor Mar 2024 – May 2025

 - Developed rover software using ROS and C++ for autonomous navigation, robotic arm control, and real-time communication
 - Designed rover networking infrastructure and mentored junior members in ROS and embedded systems

Software & Communication Team Member Oct 2022 – Mar 2024

 - Contributed to rover networking, real-time communication, and autonomous navigation testing
- MIST Computer Club** Dhaka, Bangladesh
Executive Director June 2025 – June 2026
 - Oversee all club operations, including competitive programming workshops, hackathons, and inter-university contests

- o Organized MIST Talent Hunt 2025 — coordinated competitive programming and CTF contests, managed volunteer teams and technical infrastructure

Vice President / Programming Mentor

Apr 2023 – June 2025

- o Mentored students in competitive programming, covering data structures, graph algorithms, and dynamic programming
- o Co-organized intra-university programming contests and set contest problems
- o Supported event coordination for MIST LeetCon 2023 (international cybersecurity conference) and Cyber Drill 2023

Projects

- **iinwentry – Inventory Management SaaS** Link (<https://iinwentry.com/>)

Technologies: React 19, TypeScript, Vite, Tailwind CSS, Express 5, Prisma, PostgreSQL, Supabase, Stripe Jun 2026 – Present

Developed a full-stack inventory management SaaS as a monorepo, featuring QR-coded inventory items, hierarchical stock tracking, JWT-based authentication, image uploads, transactional emails, and Stripe-powered subscription billing. Built scalable REST APIs with Express and Prisma, and implemented a responsive web interface using React, TypeScript, Vite, Tailwind CSS, and Radix UI.

- **Autonomous Trolley for Warehouse** Link (<https://raisulrahad.vercel.app/projects/autonomous-trolley>)

Technologies: ROS, OpenCV, ZED Stereo Camera, Jetson Xavier NX, Teensy, C++, Python Jun 2025 – Sep 2025

Developed an autonomous warehouse trolley using stereo vision and ROS for intelligent navigation. Implemented obstacle detection and depth estimation with ZED camera data and OpenCV, and integrated real-time communication between Jetson Xavier NX and microcontroller for motion control and system coordination.

- **MIST Mongol Barota – Robotics Team Website** Link (<https://mongol-barota.mist.ac.bd/>)

Technologies: React.js, TypeScript, Tailwind CSS, Node.js, REST API

Oct 2025 – Nov 2025

Developed and deployed the official website for MIST Mongol Barota, a competitive Mars rover team representing Bangladesh internationally. Built a responsive platform showcasing the team's rovers, competitions, achievements, research, sponsorship opportunities, and recruitment, with structured content and interactive navigation for team and public engagement.

- **PagePlay – Interactive Online Book Portal** Link (<https://github.com/Raisulll/PagePlay>)

Technologies: React.js, Tailwind CSS, Node.js, Express.js, PostgreSQL, OAuth, Chatbot API

Dec 2024 – Mar 2025

Built a full-stack web platform for eBook browsing, reviews, and user interaction. Designed RESTful APIs and relational database schemas to support scalable content management, user authentication, and chatbot-based assistance.

- **Codebook – AI-Assisted Code Learning Platform** Link (<https://github.com/Raisulll/CBook>)

Technologies: Java, JavaFX, Firebase, Supabase PgVector, Judge0 API

Nov 2023 – Dec 2023

Developed an AI-assisted platform for code sharing, execution, and learning. Integrated online code execution, semantic code search, and AI-generated explanations to support collaborative programming and self-learning.

- **AI-Powered Surveillance Robot** Link (<https://raisulrahad.vercel.app/projects/surveillance-robot>)

Technologies: ESP32-CAM, Flutter, WebSocket, Embedded Systems

Aug 2024 – Oct 2024

Developed a real-time surveillance robot with live video streaming and remote control. Implemented WebSocket-based communication between the embedded system and mobile application, along with snapshot analysis for AI-assisted user queries.

- **EyeCare – Healthcare Management Platform** Link (<https://github.com/Raisulll/EyeCare>)

Technologies: React.js, Node.js, Express.js, OracleDB, JWT, REST API

Aug 2024 – Oct 2024

Developed a role-based healthcare platform supporting appointment scheduling and data management. Implemented secure authentication using JWT and designed normalized OracleDB schemas to support transactional operations and data consistency.

Awards & Achievements

- **First Runners-Up — Anatolian Rover Challenge (Türkiye)** 2024
Achieved 2nd place among 22 finalist teams from multiple countries at METU, Ankara. Received a special award for **Best Team Structure** based on technical organization and system integration.
- **11th Place — University Rover Challenge (USA)** 2026
Led team MIST Mongol Barota to **11th place** among 38 finalist teams selected from 128 applicants worldwide at URC 2026.
- **Top 15 Worldwide — University Rover Challenge (USA)** 2025
Ranked 14th among 38 finalist teams worldwide at URC 2025. Contributed to networking, autonomous navigation, and robotic arm operation.
- **ICPC Asia Dhaka Regional Contest** 2024
Competed in the ICPC Asia Dhaka Regional Contest 2024 as team *MIST_CodeCrafters* among 300+ teams nationwide.
- **2nd Place — Independence Day Programming Contest** 2023
Achieved 2nd position among 63 teams in a 5-hour onsite programming contest focused on data structures and algorithms.

Problem Solving

- **500+ problems solved** across Codeforces (Pupil, 1290), CodeChef (3★, 1600), LeetCode (70+), and VJudge
- **Core competencies:** Arrays, Strings, Graph Traversal (BFS/DFS), Greedy Algorithms, Dynamic Programming, Recursion